Sl. No	**SECOND TERM SUMMATIVE ASSESSMENT** **STD VIII – PHYSICS ANSWER KEY** **PREPARED BY: DEEPAK M; GBHSS MALAPPURAM**	Score
1	1.c. As depth increases, liquid pressure increases.	
2	b. Both assertion and reason are true, but the reason does not explain the assertion.	
3	(a)Correct calculation: 24 × 60 × 60 = 86400 s (b)SI units are internationally accepted units; other units are derived from them.	
4 A 4 B	(a) Correct figure - X (b)The attraction and repulsion of magnets on the sides of the B mine and on the rails allows the train to stay aloft and move forward without touching the rails. Since the rails and the train can move without touching each other, they can travel at high speeds without energy loss and noise pollution due to friction. **OR** (a) P – North Pole, Q – South Pole (b) Statement is True.The assertion is true. Every magnet, however small, has two poles. A magnet with only one pole does not exist.	
5 A. 5 B	Due to various air currents in the atmosphere, positive or negative charges accumulate in clouds. At the pointed tips of the lightning conductor, charges opposite to those in the clouds are induced.The lightning conductor provides a safe path for the electric charge produced during lightning to flow into the earth safely. OR a. Do not take shelter under tall or isolated trees during lightning. Do not try to operate electrical appliances during lightning. b. A person struck by lightning should be laid flat in a well-ventilated place. Rub the entire body to provide warmth. Give artificial respiration. Press the chest firmly at regular intervals. (A note including the above points.)	
6	•Pressure = Force / Area	

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	When placed horizontally: 100 N / 1 m^2 = **100 Pa** When placed vertically: 100 N / 0.25 m^2 = **400 Pa** Horizontal: 100 Pa, Vertical: 400 Pa Greater pressure is experienced when it is placed vertically.	
7	• Car: Speed = Distance / Time = 90 m / 6 s = 15 m/s Bus: Speed = Distance / Time = 120 km / 2 hours = 120000 m / (2 × 60 × 60) = 16.67 m/s The bus travels at a higher speed.	
8 A 8 B	(a) Charging by friction Charging by contact Charging by induction (b) While taking a print, a distribution of positive charges in the shape of the letters to be photocopied is formed on the surface of the drum. The toner gives negative charge to its particles. As the drum rotates, the toner particles stick to it. Thus, an image of the document appears on the surface of the drum.The paper to be photocopied is given a strong positive charge and is passed close to the drum. As a result, the toner particles sticking to the drum are attracted to the paper and the image gets printed on the paper. **OR**	

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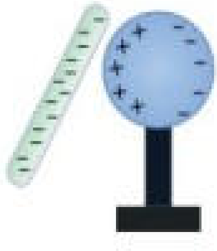
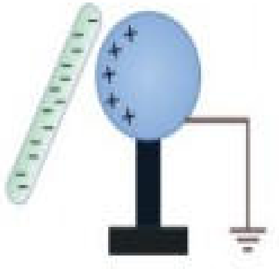
	a.  b.  When a negatively charged ebonite rod is kept near the metal sphere and the metal sphere is earthed, the negative charge present on the metal sphere is lost due to earthing.	
9	a) Rubber suckers stick to smooth surfaces because when they are pressed against the surface, the air inside escapes and the pressure inside decreases. Since the atmospheric pressure outside is greater than the reduced pressure inside, the sucker sticks firmly to the surface. (b)When heat is supplied, the air molecules inside the balloon gain more kinetic energy and move faster. This increases the pressure exerted on the walls of the balloon. As the balloon is already well inflated, it cannot withstand this additional pressure and hence bursts. (c)A compass helps to find direction. The Earth has a magnetic field. The magnetic needle of the compass always points towards the Earth’s magnetic poles (approximately in the north–south direction).Using this, navigators can identify the correct direction and travel accurately.	